

# Biases in LLM-Generated Musical Taste Profiles for Recommendation

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## Abstract

Large Language Models are increasingly used to generate user profiles in recommender systems, where a single natural-language profile is often reused both for user-facing presentation and as an internal representation for recommendation. This unified approach implicitly assumes that a profile optimized for recommendation performance is also an accurate and unbiased representation of a user’s preferences. Prior work has shown that LLM-generated profiles may introduce biases, stereotypes, or unsupported inferences, which can affect both fairness and downstream recommendation behavior.

In this work, we investigate a decoupled profile generation strategy by separating profile generation according to purpose. We generate two distinct profiles: a user-facing profile intended to provide an interpretable summary of musical taste, and a recommendation-optimized profile designed specifically for retrieval-based recommendation.

To study this idea, we analyze the effect of decoupled profile generation in a controlled recommendation setting. We compare user-facing and recommendation-optimized profiles in terms of how well they support downstream recommendation and how they differ in their representational properties. We also examine the generated profiles to assess whether stricter prompting and purpose-specific design reduce unsupported assumptions and biased descriptions.

Our results show that recommendation-optimized profiles generally improve retrieval effectiveness compared to user-facing profiles. In terms of representational properties, recommendation-optimized profiles consistently retain more interaction-supported genre information. They also slightly reduce unsupported claim rates, suggesting improved grounding. One LLM exhibits a modest increase in unsupported claims under the optimized format, but this is accompanied by the largest gains in genre coverage and retrieval performance.

Overall, the findings indicate that decoupling profile generation by purpose can improve recommendation performance while maintaining, and in some cases improving, representational faithfulness. This work contributes an empirical analysis of purpose-specific prompting for LLM-generated user profiles and provides evidence that separating presentation-oriented and recommendation-oriented representations can help balance effectiveness and bias considerations in recommender systems.

## 1 Introduction

Recommender systems play an important role in helping users discover content, especially in domains such as music streaming, where the number of available items is extremely large. Most traditional recommendation methods rely on latent representations that are learned from user–item interaction data. While these representations are usually effective for prediction, they are difficult

to interpret and provide little transparency about how user preferences are modeled.

Recently, Large Language Models (LLMs) have been proposed as a way to generate natural-language user profiles that summarize a user’s preferences in a readable and interpretable form. These profiles can describe a user’s musical taste using genres, styles, and other attributes extracted from listening history. In many recent approaches, a single LLM-generated profile is reused both as a user-facing explanation and as an internal representation for recommendation. This unified approach assumes that a profile that works well for recommendation is also an accurate and unbiased summary of the user’s taste.

However, prior research suggests that LLM-generated profiles may introduce biases, stereotypes, or unsupported inferences, especially when compressing complex user behavior into short textual descriptions. For example, a model might overgeneralize a user’s taste based on a limited set of interactions, or infer attributes that are not directly supported by the listening history. This raises concerns about fairness, faithful representation, and the potential amplification of bias in recommendation systems.

Motivated by these concerns, this work explores an alternative approach based on decoupling profile generation by purpose. Instead of generating a single profile, we produce two separate representations for each user and time window: a user-facing profile aimed at interpretability and transparency, and a recommendation-optimized profile designed specifically to support retrieval-based recommendation. The recommendation-oriented profile is generated under stricter constraints, limiting content to interaction-supported attributes and avoiding demographic or identity-related inferences. The goal is to reduce bias at the representation level while still maintaining recommendation performance.

We conduct experiments on the public Last.fm 1K listening history dataset, which is preprocessed and enriched with musical metadata. Recommendation is formulated as a retrieval task using a bi-encoder trained on user–positive–negative triplets. Performance is evaluated using standard retrieval metrics across multiple time windows. In addition, we perform a complementary bias analysis that measures how well generated profiles align with the listening evidence. Specifically, we analyze unsupported claims and genre-token coverage to examine whether decoupling improves faithfulness.

The main contributions of this work are:

1. We propose a decoupled profile generation strategy that separates user-facing and recommendation-oriented profiles.
2. We adapt a retrieval-based evaluation framework to compare the effect of different profile types on recommendation performance.

3. We introduce a faithfulness-based bias analysis that measures unsupported claims and genre coverage to assess alignment between generated profiles and listening evidence.

## 2 Background

Recent work has increasingly examined fairness and bias issues arising from the use of Large Language Models (LLMs) in recommendation systems [5]. While LLMs offer powerful tools for modeling user preferences by leveraging rich semantic knowledge, several studies show that their outputs may reflect popularity bias, stereotyping, and uneven treatment of users, raising concerns about fairness and faithful representation. These concerns are especially important in recommendation settings, where system outputs directly influence user experience and exposure to content.

Traditionally, recommender systems represent users through latent embeddings learned from interaction data, such as clicks or listening histories. Although effective, these representations are often opaque and difficult to interpret. To address this limitation, recent approaches have proposed using user profiles, and in particular natural-language user profiles, as an intermediate representation. Such profiles aim to summarize user preferences in a human-readable form, enabling greater transparency, explainability, and user trust. With the emergence of LLMs, generating these profiles automatically from user interaction data has become increasingly feasible.

However, representing users through compact textual profiles introduces new challenges. By summarizing complex and evolving user behavior into short descriptions, LLMs may overemphasize dominant or popular preferences while underrepresenting niche or temporal interests. As a result, profiles that are optimized for downstream recommendation performance may not accurately reflect how users perceive their own tastes. This creates a tension between representation quality, fairness, and recommendation utility, motivating further investigation into bias and fairness in LLM-generated user profiles for recommendation.

### 2.1 Related Work

Recent work has explored the use of natural-language user profiles as an interpretable way to represent users in recommender systems. Transparent and Scrutable Recommendations Using Natural Language User Profiles proposes generating textual profiles from user interaction data to improve transparency and user understanding [4]. While this approach increases interpretability, it also raises concerns about whether such profiles accurately reflect the full range of a user’s preferences.

Other studies focus on fairness and bias in LLM-based recommendation systems. *Is ChatGPT Fair for Recommendation?* shows that LLM-driven recommendations can favor popular items and behave inconsistently across different users [7]. Similarly, *Consumer Fairness Evaluation in Large-Language Model Recommendations* introduces evaluation methods to distinguish true personalization from stereotype-driven recommendations, emphasizing that good recommendation performance does not necessarily imply fair user representation [1].

Beyond recommendation quality, previous work has highlighted risks in using LLMs to represent people through text. *Two Tales of*

*Persona in LLMs* shows that persona-based prompting can amplify stereotypes and oversimplify individual identities [6]. In addition, *Towards Fair Large Language Model-based Recommender Systems without Costly Retraining* proposes post-hoc debiasing strategies, showing that fairness issues often originate at the representation level [3].

In contrast to prior work, we focus on bias introduced during profile generation itself. We propose decoupling user-facing profiles from recommendation-optimized profiles, allowing different representations to serve different goals. This approach addresses bias earlier in the pipeline, at the level of user representation, rather than only at the recommendation or model level.

## 3 Decoupled Profile Generation

The core idea of our work is to decouple user profile generation according to its intended purpose. In the original study, a single natural-language profile is generated for each user and then reused both as a user-facing representation and as input for recommendation evaluation. While this unified approach is convenient, it implicitly assumes that a profile optimized for recommendation performance is also a faithful and fair representation of the user’s musical taste. Prior analysis suggests that this assumption does not always hold.

In this work, we explicitly separate these two objectives. We distinguish between profiles that are generated to represent users accurately and transparently, and profiles that are generated to optimize recommendation performance. The motivation behind this design choice is that different goals may require different representations. A user-facing profile should aim to reflect the diversity and temporal evolution of a user’s listening behavior, whereas a recommendation-oriented profile may prioritize patterns that maximize retrieval accuracy, potentially amplifying popularity bias or overrepresenting dominant preferences.

By decoupling profile generation, we address fairness concerns at an earlier stage of the recommendation pipeline, namely at the level of user representation. Rather than attempting to correct bias solely through model-level debiasing or post-hoc adjustments, our approach allows profile construction itself to be adapted to different objectives. This separation enables a clearer analysis of the trade-off between representational faithfulness and recommendation effectiveness, and provides a more flexible framework for studying bias in LLM-generated user profiles.

This decoupling principle guides all subsequent stages of our method, including track sampling, prompt design, and the construction of retrieval-oriented profile representations.

### 3.1 Dataset

In this work, we aim to reproduce and extend the methodology proposed in *Biases in LLM-Generated Musical Taste Profiles for Recommendation*. However, when working with the publicly available code and data, we encountered a limitation in the evaluation stage of the original framework. In particular, some of the data required to evaluate user profiles in a recommendation setting—specifically the set of tracks embedded into the shared latent space for computing Recall and NDCG—are not included in the released dataset.

This limitation likely stems from the use of proprietary Deezer data, which prevents full reproducibility.

To address this issue and enable a controlled and transparent experimental setup, we instead rely on the publicly available Last.fm 1K dataset. This dataset provides detailed user listening histories, including user identifiers, timestamps, artists, and track information, and allows us to reconstruct a recommendation scenario similar to that of the original study. Using this dataset, we are able to generate user profiles, sample positive and negative items, and evaluate recommendation performance without relying on unavailable internal data. After selecting the dataset, we preprocess and adapt it to follow the same principles as the original methodology, including user selection, time-window segmentation, and sampling of tracks for profile generation, as described in the following sections.

### 3.2 Preprocessing

Because the original Deezer dataset used in the target paper is not fully available, we base our experiments on the public Last.fm 1K listening history dataset. Each record contains a user identifier, timestamp, and track–artist information. We preprocess the raw logs to align with the profile-generation and evaluation setting of the original study.

First, we restrict the data to a single calendar year (2009). Listening activity in the dataset spans multiple years but is highly imbalanced, with earlier years containing substantially fewer interactions. Focusing on the most data-rich year ensures sufficient coverage for all users and enables consistent construction of time windows (30, 90, 180, and 365 days).

Second, we clean the textual fields by removing entries with corrupted or unreadable artist or track names (e.g., encoding artifacts). Such entries cannot be reliably matched to external metadata sources and would otherwise introduce noise into the profile-generation prompts.

After preprocessing, we retain listening data for 83 users. For the main experiments, we focus on a subset of 50 users, matching the scale of the original study. The remaining users are retained to support downstream sampling procedures, such as providing additional tracks for negative examples during evaluation.

Finally, we enrich each track with additional metadata required for prompting the LLM, including playcount, release year, artist country of origin, main genre, and secondary genre. While playcount is available directly in the logs, the remaining attributes are retrieved from external music databases via programmatic queries with caching and rate limiting. Tracks for which reliable metadata cannot be obtained are excluded from subsequent steps. The resulting dataset provides structured textual descriptions for each track, used as input to the LLM in the profile-generation stage.

### 3.3 Time Window Construction

To capture both short-term and long-term user preferences, we segment each user’s listening history into multiple time windows. Following the methodology of the original study, we consider four time windows of increasing length: 30, 90, 180, and 365 days. All time windows start from the beginning of the selected year, January 1st, 2009, and extend forward in time according to their respective durations.

The motivation behind using multiple time windows is to reflect the evolving nature of musical preferences. Shorter windows (30 and 90 days) capture more recent and potentially unstable listening behavior, while longer windows (180 and 365 days) reflect more stable and long-term preferences. This design allows us to study how the representativeness and effectiveness of LLM-generated profiles change as more historical data is included.

For each user, we construct a separate listening subset for each time window. These subsets are treated independently in subsequent steps, meaning that each user can have multiple profiles corresponding to different time horizons. This setup is consistent with the original study and enables a direct comparison across time windows when evaluating profile quality and recommendation performance.

### 3.4 Track Sampling

For each user and each time window (30, 90, 180, and 365 days), we select a fixed set of 15 tracks that will be used as evidence for prompting the LLM and generating the user profile. The goal of this sampling step is twofold: (1) provide enough coverage to reflect the user’s taste within the window, and (2) keep the prompt short enough to fit within a single-shot context for all LLMs.

**3.4.1 Candidate interactions per user–window.** For a given user  $u$  and a time window  $w$ , we first collect all listening events that fall inside the window (starting from January 1st, 2009). Each event contains the user ID, timestamp, artist name, and track name. This produces a set of candidate tracks listened to by the user in that specific period.

**3.4.2 Aggregation of play frequency.** To rank items by importance within a window, we aggregate listening frequency. Specifically, we compute a playcount proxy per user–window by counting how many times each (artist, track) pair appears in the logs within that window. If a track occurs multiple times, it receives a higher playcount, reflecting stronger engagement. This aggregated value is later included in the metadata shown to the LLM.

**3.4.3 Two-step sampling.** Directly selecting the top 15 most-played tracks can lead to a highly redundant set dominated by a small number of artists (e.g., a user repeatedly streaming a single album). To avoid this and obtain broader coverage of a user’s taste, we apply a two-step strategy inspired by the original study. Step 1 - Select top artists: For each user–window, we rank artists by the total number of plays in that window. We then select the top 15 artists as the primary preference signals. Step 2 - Select one representative track per artist: For each of the selected artists, we choose one track to represent that artist in the profile evidence. Concretely, we pick the artist’s most-played track within that time window. This ensures that each of the 15 sampled items corresponds to a different artist and that the selected track is the most representative choice for that artist in that specific period.

**3.4.4 Missing data.** In practice, some user–window pairs may contain insufficient listening activity after year filtering and cleaning. For example, a user might listen to fewer than 15 distinct artists during a short window (e.g., 30 days), or some tracks may be removed during preprocessing due to missing metadata or corrupted

text fields. In these cases, the sampling procedure returns fewer than 15 items. To maintain consistency for downstream prompting, we restrict later stages to user–window pairs that yield exactly 15 sampled tracks.

**3.4.5 Output format.** The final result of this sampling step is a structured table for each user–window containing 15 rows, each row describing a sampled track with its associated metadata fields (e.g., artist name, track title, playcount, release year, country of origin, and genres). This table is later converted into a textual list and inserted into the LLM prompt in Section 3.5.

### 3.5 Profile Generation with LLMs

For each user–time window pair, we generate a textual music preference profile using a large language model (LLM), based solely on the 15 sampled tracks obtained in the previous step. Each profile is generated independently per user–window and is conditioned only on the associated track metadata, without incorporating any external user information.

The input to the LLM consists of a structured textual list of the 15 sampled tracks, where each track is represented by its metadata fields, including track title, artist name, playcount within the time window, release year, country of origin, and genre information. This metadata is inserted verbatim into a fixed prompt template to ensure consistency across users and windows and to avoid introducing information not supported by the data. We generate profiles using multiple LLMs in order to study the effect of the underlying model on the resulting representations. Specifically, we use Gemini [2], accessed via an external API, as well as DeepSeek and LLaMA, which are deployed and executed locally. For all models, identical prompt templates and output format constraints are used to ensure comparability across LLMs.

For each user–window pair, we generate two complementary profile types. The first is a user-facing profile, expressed as a short natural-language paragraph summarizing the user’s apparent musical preferences, such as favored genres, styles, or eras, strictly grounded in the provided tracks. The second is a recommendation-optimized profile, designed to support retrieval-based recommendation models. This profile consists of a structured list of preference keywords (e.g., genres, moods, production styles) and a small set of concise query-style sentences that can be directly used for matching against item representations.

To reduce variability and hallucination, all prompts enforce strict formatting constraints and explicitly instruct the model not to infer demographic attributes or preferences not evidenced by the input tracks. Profile generation is performed deterministically per prompt, and each generated profile is assigned a unique identifier corresponding to its user and time window. These profiles serve as query representations in the downstream recommendation experiments described in the following sections.

**3.5.1 Decoupled Profiles for Recommendation.** Following the decoupling principle introduced in Section 3, we operationalize recommendation oriented profiles using a structured output format tailored to retrieval-based models. Unlike user-facing profiles, which are expressed as free-form natural language, recommendation-oriented profiles are generated in a constrained representation

designed to align closely with item metadata and dense retrieval objectives.

Each recommendation-oriented profile consists of two components: a set of explicit preference keywords and a small number of short query-style sentences. The keyword component captures salient attributes such as genres, subgenres, eras, countries of origin, moods, instruments, and production characteristics that are directly comparable to item metadata fields. This explicit enumeration of attributes increases lexical and semantic overlap between profile representations and track descriptions in the shared embedding space. The query-style sentences provide a complementary abstraction level, expressing preferences as concise, intent-like statements. These sentences are intentionally short and focused, mirroring the structure of queries commonly used to train sentence embedding models. Compared to longer narrative descriptions, such formulations reduce ambiguity and emphasize discriminative signals relevant for retrieval.

To ensure consistency and robustness across users and models, the prompt enforces strict constraints on output length, structure, and content. Profiles are restricted to information supported by the sampled tracks, and demographic or identity-based inferences are explicitly disallowed. These constraints limit the introduction of noise and bias and encourage the model to compress interaction evidence into a stable, retrieval-efficient representation. This structured formulation enables the language model to produce profiles that are directly compatible with bi-encoder architectures, facilitating effective similarity-based matching between user representations and candidate items in downstream recommendation tasks. We next describe how the generated profiles are used to construct training and evaluation datasets for retrieval-based recommendation.

## 4 Evaluation

We evaluate the proposed decoupled profile representations in a retrieval-based recommendation setting using explicitly constructed training and test datasets. Both datasets are derived from the cleaned user listening logs and the profiles generated in Section 3, and are designed to ensure a fair and leakage-free comparison between profile types.

### 4.1 Training and Test Set

**Training Set Construction.** Training instances are formulated as user–positive–negative triplets, following standard practice in bi-encoder–based retrieval models. Each triplet consists of a profile representation (serving as the query), a positive track, and a negative track. For each user–time window pair, the sampled tracks used for profile generation (Section 3.5) serve as positive examples. To construct negatives, we first define a global pool of 1000 candidate negative tracks. The size of this global negative pool is deliberately bounded to limit the number of distinct tracks requiring metadata enrichment, which proved computationally expensive when applied at larger scale. By pairing each positive interaction with multiple negatives sampled from this pool, we preserve strong contrastive training signals. All profile types are trained using the same negative pool, ensuring a fair and controlled comparison. Tracks that

were used for profile generation for any user or time window are excluded globally from this pool to prevent information leakage.

For each positive track associated with a given user–window pair, we sample a fixed number of negative tracks from the global pool, subject to the constraint that the target user has not interacted with the negative track. This ensures that negatives represent plausible but unseen items for the user. The same set of negative tracks is used across different profile types (user-facing and recommendation-optimized) to enable a controlled comparison. Each training triplet therefore corresponds to a single positive interaction paired with multiple negative examples, enabling efficient contrastive training of retrieval models.

**Test Set Construction.** The test set is constructed independently of the training data and is used to evaluate ranking performance. For each user–time window query, we define a candidate set consisting of positive and negative tracks.

Positive test examples are selected from tracks that the user has interacted with during the evaluation period but were not used for profile generation, ensuring that test positives are unseen during training. Negative test examples are sampled from a separate global pool of tracks that is disjoint from the training negative pool and also excludes all profile-generation tracks. As in training, negatives are filtered to ensure that the user has not interacted with them.

For each query, we sample a fixed number of positive and negative candidates, forming a ranking task in which the retrieval model must distinguish relevant items from non-relevant ones. The same test candidate sets are used for all profile types, guaranteeing that performance differences can be attributed solely to the profile representations rather than differences in evaluation data.

## 4.2 Experimental Setup

We evaluate the quality of the generated user profiles using a retrieval-based recommendation setting. Following prior work, we adopt a bi-encoder architecture in which user profiles and track metadata are embedded independently into a shared latent space, and relevance is determined via cosine similarity between their embeddings.

For training, we use all generated profiles as queries. For each user–time window pair, the tracks sampled during the profile generation stage serve as positive examples, while negative examples are drawn from tracks the user did not interact with in the corresponding year. Training instances are formulated as user–positive–negative triplets. Negative tracks are sampled from a fixed pool of 1,000 candidate tracks drawn from the cleaned listening logs, with all tracks used for profile generation excluded to prevent information leakage. The bi-encoder is trained for a single epoch using the AdamW optimizer, with a batch size of 10 and a maximum sequence length of 512 tokens.

For evaluation, we follow the protocol described in the original study. For each user, we sample 10 tracks the user has interacted with but which were not used for profile generation, and combine them with 500 negative tracks drawn from a disjoint negative pool, ensuring that none of the negatives were seen during training for that user. Each profile is used as a query to retrieve and rank the candidate tracks based on cosine similarity in the embedding space.

We report Recall@10 and NDCG@10 as evaluation metrics. Results are averaged over five runs with different random seeds to account for randomness in negative sampling and model initialization. All experiments are conducted using the same bi-encoder architecture and candidate sets across profile types and language models, ensuring a fair and controlled comparison. The quantitative recommendation performance obtained under this setup is presented and analyzed in the Results section.

## 5 Bias Analysis

Although our main evaluation focuses on retrieval performance, the motivation behind decoupling profile generation is strongly related to concerns about bias in LLM-generated user representations. Previous research has shown that language models can sometimes produce summaries that include assumptions or generalizations that are not fully supported by the underlying data. Even when recommendation accuracy is high, the way a user is described may still contain distortions.

In the original study, bias was evaluated using user feedback on the generated profiles. Since we do not have access to human annotations, we adopt a different approach that focuses on how well each profile aligns with the listening evidence used to generate it. More specifically, for each user and time window, we compare the generated profile to the 15 sampled tracks that were provided to the model as input.

The following subsection describes the exact metrics used to quantify this alignment.

### 5.1 Experimental Setup

We define representational bias as the inclusion of claims that are not supported by the user’s listening history. To measure this, we compute an unsupported-claim rate, which represents the proportion of extracted claims in the profile that do not appear in the corresponding evidence tracks. In addition, we calculate genre-token coverage, which reflects how much of the genre information present in the sampled tracks is actually captured in the profile.

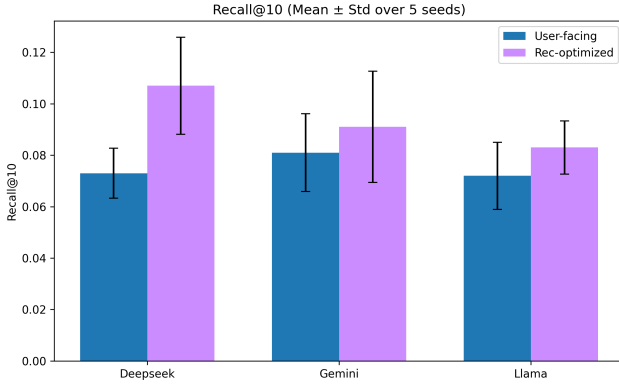
This analysis allows us to examine whether recommendation-optimized profiles remain grounded in the listening data and whether the decoupling strategy helps reduce unsupported or exaggerated descriptions of users.

## 6 Results

### 6.1 Recommendation Performance

**Table 1: Recommendation performance (mean  $\pm$  std over 5 seeds).**

| Model    | Profile       | Recall@10         | NDCG@10           |
|----------|---------------|-------------------|-------------------|
| Gemini   | User-facing   | 0.081 $\pm$ 0.015 | 0.035 $\pm$ 0.007 |
| Gemini   | Rec-optimized | 0.091 $\pm$ 0.022 | 0.051 $\pm$ 0.010 |
| Llama    | User-facing   | 0.072 $\pm$ 0.013 | 0.034 $\pm$ 0.006 |
| Llama    | Rec-optimized | 0.083 $\pm$ 0.010 | 0.033 $\pm$ 0.004 |
| Deepseek | User-facing   | 0.073 $\pm$ 0.010 | 0.033 $\pm$ 0.005 |
| Deepseek | Rec-optimized | 0.107 $\pm$ 0.019 | 0.049 $\pm$ 0.010 |



**Figure 1: Recall@10 (mean ± std over 5 seeds).**

Table 1 reports the recommendation results in terms of Recall@10 and NDCG@10 (mean ± std over five seeds). Overall, the differences between profile types are moderate but consistent for some models.

For Gemini, recommendation-optimized profiles slightly improve performance compared to user-facing profiles. Recall@10 increases from  $0.081 \pm 0.015$  to  $0.091 \pm 0.022$ , and NDCG@10 improves from  $0.035 \pm 0.007$  to  $0.051 \pm 0.010$ . A similar trend is observed for DeepSeek, where recommendation-optimized profiles achieve higher Recall@10 ( $0.107 \pm 0.019$  vs.  $0.073 \pm 0.010$ ) and higher NDCG@10 ( $0.049 \pm 0.010$  vs.  $0.033 \pm 0.005$ ). This suggests that the structured recommendation-oriented format helps the model better align profiles with relevant items during ranking.

For LLaMA, the improvement is smaller but still visible. Recall@10 increases from  $0.072 \pm 0.013$  to  $0.083 \pm 0.010$ , while NDCG@10 slightly decreases from  $0.034 \pm 0.006$  to  $0.033 \pm 0.004$ , indicating that gains in retrieval may not always translate into improved ranking quality.

Across models, DeepSeek with recommendation-optimized profiles achieves the highest Recall@10 overall (0.107), while Gemini achieves the highest NDCG@10 (0.051). The relatively small standard deviations across seeds indicate stable training behavior.

Overall, these results suggest that decoupling the profile into a recommendation-optimized format generally improves retrieval effectiveness, particularly for Gemini and DeepSeek, without introducing instability across runs.

## 6.2 Bias Analysis

**Table 2: Faithfulness-based bias analysis aggregated across all time windows.**

| Model    | Profile       | Unsupported | Coverage | Claims |
|----------|---------------|-------------|----------|--------|
| Gemini   | User-facing   | 0.216       | 0.265    | 13.515 |
| Gemini   | Rec-optimized | 0.199       | 0.574    | 32.155 |
| Llama    | User-facing   | 0.281       | 0.188    | 9.930  |
| Llama    | Rec-optimized | 0.271       | 0.330    | 16.500 |
| Deepseek | User-facing   | 0.217       | 0.245    | 11.730 |
| Deepseek | Rec-optimized | 0.262       | 0.666    | 35.115 |

Table 2 presents the results aggregated across all time windows.

For Gemini and LLaMA, recommendation-optimized profiles show slightly lower unsupported-claim rates compared to user-facing profiles. In Gemini, the rate decreases from 0.216 (user-facing) to 0.199 (recommendation-optimized). For LLaMA, it decreases from 0.281 to 0.271. The differences are not very large, but they suggest that structuring the profile for recommendation does not introduce more unsupported claims and may even reduce them slightly.

DeepSeek shows a different pattern. In this case, the unsupported-claim rate increases from 0.217 in user-facing profiles to 0.262 in recommendation-optimized profiles. This means that, for DeepSeek, the recommendation-oriented format may include more statements that are not directly grounded in the listening evidence. However, the increase is moderate and does not indicate extreme distortion.

When looking at genre-token coverage, the pattern is more consistent across all three models. Recommendation-optimized profiles retain much more genre-related vocabulary that appears in the sampled tracks. For Gemini, coverage increases from 0.265 to 0.574. For LLaMA, it increases from 0.188 to 0.330. DeepSeek shows the strongest increase, from 0.245 to 0.666. This suggests that the structured recommendation format captures more of the genre information that is actually supported by the listening history.

Claims refer to explicit references to genres, artists, countries, or years that can be matched against the evidence vocabulary. As expected, recommendation-optimized profiles contain substantially more claims than user-facing profiles. For Gemini, the average number of claims increases from 13.52 (user-facing) to 32.16 (rec-optimized). For LLaMA, it increases from 9.93 to 16.50. DeepSeek shows the largest increase, from 11.73 to 35.12 claims. However, a higher number of claims does not automatically imply higher bias. Bias is better reflected by the unsupported claim rate, which measures how many of these statements are not grounded in the user’s listening history.

Overall, these results show that the effect of decoupling depends on the model. For Gemini and LLaMA, recommendation-optimized profiles appear slightly more grounded. For DeepSeek, there is a small increase in unsupported claims, but at the same time, genre coverage improves substantially. This analysis complements the retrieval results by focusing on how well the profiles reflect the user’s actual listening behavior, rather than only how well they perform in ranking.

## 7 Discussion

The results show that changing the profile format has a noticeable effect on recommendation performance. In most cases, the recommendation-optimized profiles lead to better Recall@10 compared to the user-facing versions. This is especially clear for Gemini and DeepSeek. It seems that when the profile is structured in a more task-oriented way, the model can better match users with relevant tracks.

At the same time, improving recommendation performance does not automatically mean increasing bias. For Gemini and LLaMA, the unsupported claim rate actually decreases slightly when using recommendation-optimized profiles. This suggests that restructuring the profile does not harm faithfulness, and may even help keep

the content grounded in the listening history. In addition, genre-token coverage increases significantly for all three models, which means that the optimized profiles preserve more genre information that actually appears in the user’s track data.

DeepSeek shows a slightly different behavior. Although it achieves the highest Recall@10 when using recommendation-optimized profiles, it also shows a moderate increase in unsupported claim rate. This means that the model sometimes introduces more statements that are not directly supported by the listening evidence. However, the increase is not extreme, and the large improvement in genre coverage suggests that most of the added information is still relevant to the user’s music preferences.

Another important point is stability. The relatively small standard deviations across seeds indicate that the results are consistent and not highly sensitive to random initialization. This strengthens the reliability of the findings.

Overall, the effect of decoupling depends on the model. For some models, it improves both recommendation performance and grounding. For others, it improves retrieval but slightly increases unsupported claims. This shows that the interaction between prompt structure and model behavior is important.

## 8 Conclusion

This project examined the effect of decoupling user profiles into two formats: user-facing summaries and recommendation-optimized representations. We evaluated both formats along two dimensions: recommendation performance (Recall@10 and NDCG@10) and representational faithfulness (unsupported claim rate and genre-token coverage).

Our findings show that recommendation-optimized profiles generally improve retrieval effectiveness, particularly for Gemini and DeepSeek. At the same time, for most models, this restructuring does not increase unsupported claims and may even reduce them. Although DeepSeek exhibits a slight increase in unsupported claims under the optimized format, it also achieves the strongest improvements in genre coverage and retrieval performance.

Overall, the results suggest that separating the communicative goal of user summaries from the optimization goal of recommendation can be beneficial. Structured, evidence-aware prompts can enhance alignment with item features while maintaining reasonable faithfulness to user listening behavior.

### 8.1 Future Work

Several directions for future research emerge from this study. First, additional evaluation across larger user sets and longer time windows would help assess scalability and robustness. Second, alternative profile generation strategies (e.g., constrained decoding or evidence citation mechanisms) could further reduce unsupported claims. Finally, extending the analysis to other recommendation architectures, including cross-encoders or hybrid models, may provide deeper insights into how profile structure interacts with retrieval mechanisms.

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